

CHIRANJEEVI VANTAKU

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EDUCATION

Stevens Institute of Technology, Hoboken, New Jersey

Sep 2024 - May 2026

Master of Science in Computer Science

GPA: 3.9/4.00

Relevant skills: Large Language Models, SQL, Deep Learning, NLP, Data Analysis, Modeling, Computer Vision, Fine-tuning, Databricks, Problem-Solving, Optimization

WORK EXPERIENCE

Medbridge Inc., United States

May 2025 - Present

AI Developer Intern

- Architected a **multi-agent** healthcare automation system with 6 specialized AI agents handling clinical documentation, benefits verification, prior authorization, fax processing and billing - reducing administrative workload by 40+ hours per week and processing documents with **97%+ accuracy**.
- Developed an AI-powered ambient scribe that listens to patient encounters in real-time and generates structured, codified SOAP notes with automatic CPT/ICD mapping, eliminating after-hours charting for clinicians.
- Engineered an **intelligent document automation pipeline** using OCR and medical NLU to classify and extract data from **50+ medical document types** including referrals, lab reports, EOBs, and insurance forms, integrating seamlessly with **Epic, Cerner** and **150+ EHR systems** via **HIPAA-compliant HL7/FHIR** standards.
- Built a **Voice AI agent** for automated insurance eligibility verification and appointment scheduling, processing up to 100 eligibility checks per minute through secure payer portal integrations with Availity and Waystar, reducing claim denials.

Beats by Dre, United States

Dec 2024 - Mar 2025

Data Analyst Intern

- Engineered targeted consumer insights frameworks using **AIDA methodology**, building data-driven customer journey maps and segment-specific "Ideal Consumer Lists" from **50K+ survey responses** that directly shaped strategic marketing decisions and campaign targeting.
- Conducted in-depth quantitative and qualitative data analysis on survey responses, leveraging Python, SQL, and **AI-powered analytics tools** for advanced visualizations, uncovering behavioral patterns that drove a **22% increase in product engagement**.
- Designed and delivered **AI-augmented dashboards and executive presentations** using tools like Claude and Gamma, reducing stakeholder **reporting time by 60%** and optimizing data-driven decision-making across product strategy teams.

Columbia University, United States

Sep 2024 - Nov 2024

Research Assistant

- Investigated **MM-Diffusion architecture** to generate synchronized high-fidelity audio-video content by leveraging novel denoising and Random-Shift-based Multi-Modal Attention mechanisms, enhancing cross-modal alignment and coherence while reducing computational redundancy.
- Published research findings showing 28% better FVD scores and 31% improved FAD metrics on AIST++ dataset, outperforming 3 state-of-the-art baseline models.
- Optimized multi-modal attention mechanism that **decreased training time by 50%** while maintaining 94% generation quality, potentially saving **\$15K in GPU costs**.

PUBLICATIONS

High-Dimensional Audio Genre Classification Using MFCC-Driven Multi-Layer Perceptrons.

[\[Paper \]](#)

- Published in IRJET, Volume 12, Issue 05 | May 2025. Developed a deep learning-based music genre classifier using MFCCs and MLPs, achieving **64% validation accuracy** on the GTZAN dataset through dimensionality reduction and regularization techniques.

PROJECTS

FinSight: AI-Powered Multi-Agent Stock Research Assistant | *LangChain, LangGraph, RAG, FAISS, Streamlit*

- Built a **multi-agent research assistant** using LangGraph with specialized agents for SEC filing retrieval, sentiment analysis of financial news, and technical indicator computation, leveraging **RAG with FAISS vector store** over 10,000+ earnings reports to deliver real-time investment insights through an interactive **Streamlit dashboard** with **85%+ relevance accuracy**.

MedScribe: Clinical Note Summarizer with Fine-Tuned LLM | *PyTorch, LoRA, HuggingFace, FastAPI, Docker*

- Fine-tuned a **LLaMA-based model using LoRA** on a public clinical note's dataset (MIMIC-III) to generate structured **SOAP note summaries** from unstructured medical text, deployed as a **Dockerized FastAPI microservice** with MLFlow experiment tracking, achieving **87% ROUGE-L score** and reducing summarization time by **70%** compared to manual review.

SKILLS

Languages: Python, C, SQL, MATLAB, Arduino

AI Technologies: HuggingFace, LangChain, LangGraph, LangSmith, LlamaIndex, vLLM, Unsloth, RAG Pipelines, Vector Databases, LLMs, Multi-Agent Orchestration, MCP, Transformers.

Packages: PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, OpenCV, SpaCy, EasyOCR, Tesseract.

Data & Analytics: Power BI, Excel, Data Visualization, Regression Analysis, Clustering, Statistical Methods, A/B Testing.

Cloud & DevOps: Microsoft Azure, GCP, Vertex AI, Databricks, Docker, Kubernetes, MLFlow, Flask, Streamlit, REST APIs, CI/CD Pipelines, GitHub.

Core Competencies: Generative AI, Fine-tuning (LoRA/QLoRA), Prompt Engineering, Agentic Workflow Automation, OCR, Deep Learning, NLP, Computer Vision, Predictive Modeling, MLOps, Model Monitoring, Data Modeling, Agile Methodologies.

Developer Tools: Claude Code, GitHub CoPilot, Cursor AI, CrewAI, n8n.

Certifications: Microsoft Azure AI (AI-900), Microsoft Security Compliance and Identity (SC-900).